

Sr. No.	Content
1	Problem statement & Problem definition
2	Abstract
3	Introduction
4	Existing system
5	Objective
6	Literature review
7	Methodology
8	Component
9	Technical requirements
10	Future scope
11	Result
12	Conclusion
13	Refrences

Problem Statement

In regions with limited access to energy-efficient cooling systems, how can students design and optimize a thermoelectric Peltier - based cooling device that utilizes waste heat or renewable energy sources, applying thermodynamics and sustainable engineering principles to enhance efficiency and reduce environmental impact?

Problem Definition

The growing need for sustainable and energy-efficient cooling technologies calls for alternatives to conventional systems that consume high power and use harmful refrigerants. Thermoelectric Peltier modules offer a solid-state, eco-friendly solution capable of providing localized cooling and waste heat recovery. This project focuses on designing and optimizing a Peltier-based cooling system powered by renewable or waste energy sources to achieve efficient and affordable temperature control. By applying thermodynamic principles and optimizing heat transfer performance, the project promotes green engineering practices and supports the theme “Empowering Quality Engineering Education through Applied Mechanics and Sustainability.”

Abstract

Conventional refrigeration processes release ammonia and freon into the atmosphere, which results in global warming. These problems may be overcome by using thermoelectric modules because of the absence of coolants or refrigerants in these systems. However, the cooling performances of such modules are relatively small in comparison to those of conventional refrigerators. In this paper, the working principles of thermoelectric modules are discussed together with a review of different relevant aspects, namely: the thermoelectric materials, and their mechanical properties used to build thermoelectric devices, different types of thermoelectric devices available on the market, mathematical modelling of thermoelectric materials, and various applications of thermoelectric materials in different fields.

Introduction

The rising global energy demand and overuse of non-renewable resources have led to severe environmental issues. Transitioning to renewable and green technologies is essential, and thermoelectric devices offer a clean, sustainable way to convert waste heat into useful energy. Thermoelectricity was first observed by Thomas Seabek (1821), who discovered that a temperature difference between two dissimilar metals can generate an electric current. Later, Jean Charles Athanase Peltier (1834) found that passing current through two different materials causes heat absorption or release at their junctions — known as the Peltier effect. Thermoelectric devices, based on these effects, are solid-state systems that directly convert heat into electricity. They are compact, eco-friendly, maintenance-free, and have no moving parts, making them reliable for energy conversion and cooling applications. Peltier Effect & Thermoelectric Principle: Jean Charles Athanase Peltier showed that when electric current passes through two different materials, heat is absorbed at one junction and released at the other. This effect is commonly observed in N- type and P-type thermoelectric materials.

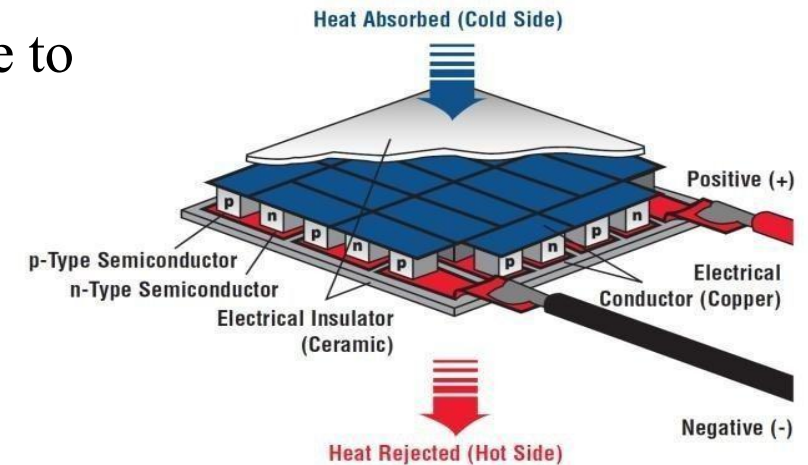
Existing System

Conventional Refrigeration System

- Use compressors, refrigerants (like Freon, ammonia), and moving parts to transfer heat.
- Efficient for large-scale cooling but **environmentally harmful** due to greenhouse gas emissions.
- Require regular maintenance and have mechanical wear over time.

Limitations of Existing Systems:

- Release harmful gases contributing to global warming and ozone depletion.
- Bulky and not suitable for small-scale or portable applications.
- Mechanical parts increase chances of failure and require frequent maintenance.



Objective

1. To design and develop a thermoelectric cooling system using a Peltier module (TEC 12706).
2. To demonstrate the Peltier effect practically by creating a temperature difference between hot and cold sides.
3. To monitor and record temperature variations using a temperature sensor and Arduino interface.
4. To analyze the performance of the Peltier module under different input voltages and cooling conditions.
5. To explore a compact, energy-efficient, and eco-friendly alternative to conventional refrigeration systems.
6. To study the effect of heat dissipation (with heatsink and fan) on the cooling efficiency of the system.

Literature Review

Thermoelectric cooling is an emerging, eco-friendly alternative to conventional refrigeration systems, as it eliminates the need for harmful refrigerants such as ammonia and chlorofluorocarbons (CFCs). This technology is based on the **Peltier effect**, where heat transfer occurs due to the flow of electric current across two dissimilar materials, enabling solid-state cooling.

However, several limitations have been reported in existing studies. Thermoelectric systems generally exhibit lower cooling efficiency compared to traditional vapor-compression systems. In addition, their practical applications are restricted due to inadequate heat dissipation on the hot side. The overall performance of Peltier devices is highly sensitive to input voltage, current, material properties, and thermal management techniques.

This project focuses on improving the performance of thermoelectric cooling systems through effective heat management and optimization techniques. Enhanced heat dissipation is achieved by incorporating efficient heat sinks on the hot side. The study also experimentally investigates the influence of input voltage and current on cooling performance to determine optimal operating conditions.

METHODOLOGY

Selection of Components

Chose a Peltier thermoelectric module suitable for small-scale cooling experiments. Selected compatible heat sinks for effective heat dissipation on the hot side.

Used temperature sensors (like thermocouples or digital sensors) to measure surface temperatures accurately.

Prepared a regulated DC power supply to provide controlled voltage and current to the module.

Setup Assembly

Mounted the Peltier module securely between two heat sinks, ensuring good thermal contact using thermal paste. Connected temperature sensors on both the hot and cold sides of the module for real-time monitoring.

Wired the module to the power supply with appropriate connectors and ensured stable electrical connections.

Experimental Procedure

Powered the Peltier module with incremental voltage steps to observe the change in temperature on both sides. Recorded the temperature readings at each voltage/current level after allowing the system to stabilize.

Monitored the cooling effect on the cold side and the heat dissipation on the hot side during operation. Ensured consistent environmental conditions to minimize external effects on temperature measurements.

Safety and Maintenance

Used proper insulation and handled the module carefully to avoid damage from overheating or electrical short circuits. Regularly checked connections and ensured the heat sinks were not clogged or obstructed.

COMPONENTS

- Peltier Module (TEC 12706): Core thermoelectric device for heat transfer.
- Cooling Master Heatsink (H410R): For effective heat dissipation on the hot side.
- PWM Cooling Fan: Improves airflow and enhances cooling performance.
- Temperature Sensor (e.g., LM35/DS18B20): Measures cold and hot side temperatures.
- Arduino Board: Reads temperature data and sends it to the computer.
- DC Power Supply (12V): Provides controlled power input to the Peltier module.
- Resistor (4.7 k Ω): Used for voltage divider in temperature sensor circuit.
- 12V Temperature Display: Shows real-time temperature readings.

Technical Requirements

1. Hardware Requirements

- **Peltier Module** (TEC1-12706): for thermoelectric cooling
- **Heat sinks** (hot side and cold side) for effective heat dissipation
- **DC power supply** (0–15V, 0–10A) to provide controlled input
- **Cooling fan** (s) for forced convection on the hot side
- **Thermal paste** to improve heat transfer between surfaces
- **Temperature sensors** (e.g., thermocouples or digital sensors)
- **Arduino Board:** Processes data from sensors and transmits it for analysis and display.

2. Software Requirements

- **Arduino IDE:** Used for writing, compiling, and uploading code to the Arduino board. It enables communication between hardware components
- **Serial Monitor / Serial Plotter:** Built-in tools in Arduino IDE used to observe real-time temperature data and analyze system performance.
- **Programming Language:** Embedded C/C++ is used for coding the Arduino to control sensors and process data.

3. Libraries Used

- **OneWire Library:** Enables communication with DS18B20 temperature sensor using one-wire protocol
- **DallasTemperature Library:** Simplifies temperature data reading from DS18B20 sensor.
- **LiquidCrystal / LiquidCrystal_I2C Library:** Used to interface and display temperature readings on the display unit.

Future Scope

Improving Efficiency:

Use advanced thermoelectric materials with higher Seebeck coefficients to enhance performance.
Implement multi-stage (cascade) Peltier modules for greater temperature differences.

Smart Control Systems:

Integrate microcontrollers with PID algorithms for automatic temperature control. Develop IoT-based monitoring systems for real-time data tracking and remote operation.

Enhanced Cooling Design:

Optimize heat sink and fan configurations for better heat dissipation and compact design.
Employ liquid cooling or hybrid systems to improve efficiency for high-load applications.

Power Optimization:

Combine Peltier systems with renewable power sources such as solar energy for sustainable cooling.

Applications Expansion:

Use in portable refrigeration, CPU cooling, medical sample storage, and space electronics.

Result

The thermoelectric Peltier module successfully demonstrated the conversion of electrical energy into a temperature difference across its two surfaces. Upon applying DC current, one side of the module cooled significantly while the opposite side heated up, confirming the Peltier effect. The temperature difference increased with higher current, reaching a maximum ΔT of around 40–60°C under optimal conditions. Reversing the current polarity switched the hot and cold sides, illustrating the reversible nature of the effect. The efficiency of cooling was strongly dependent on effective heat dissipation using heat sinks and fans. Overall, the experiment validates the practical application of Peltier modules in compact and portable cooling devices.

Conclusion

The project successfully demonstrates the working principle of the Peltier effect, where electrical energy is used to create a temperature difference. A compact and eco-friendly cooling system was developed without the use of harmful refrigerants. The temperature variation data collected using Arduino and sensors validated the performance of the Peltier module under different voltage conditions. Use of efficient heat dissipation (heatsink and fan) improved cooling stability and module lifespan. Though less efficient than conventional refrigeration, the system shows great potential for small-scale and portable cooling applications. This experiment lays the foundation for future improvements in thermoelectric efficiency and smart cooling technologies.

References

1. Jean Charles Athanase Peltier (1834). *On a new phenomenon of electricity*. Annales de Chimie et de Physique.
2. D. M. Rowe (Ed.). (2006). *Thermoelectrics Handbook: Macro to Nano*. CRC Press, Taylor & Francis Group.
3. H. Julian Goldsmid (2010) *Introduction to Thermoelectricity*. Springer Series in Materials Science.
4. U. S. Department of Energy. (2023). *Thermoelectric Energy Conversion Technology Overview*. Available at: <https://www.energy.gov>
5. NASA. (2020). *Thermoelectric Devices for Space Applications*.
6. ScienceDirect. Research papers on thermoelectric modules and Peltier cooling systems
7. International Energy Agency (2022). *Advances in Thermoelectric Cooling Systems*.
8. IEEE Transactions on Components Packaging and Manufacturing Technology Various articles on thermoelectric cooling modules.
9. Electronics Cooling Magazine. Various articles on thermoelectric cooling modules.
10. MIT OpenCourseWare. Materials on thermoelectric effects and heat transfer.

Thank You